Question 1:

a) if (grade >= 90)
{
system.out.println(“Great Job!”);
}

b) if (number < 20 II number > 50)
{
system.out.println (“Error”);
}

c) if (y < 100)
{
y=y+2;
}

Question 2:

```
If (num1 > num2)
{
system.out.println(“First number is larger”);
}
else if (num2 > num1)
{
system .out.println(“Second number is larger”);
}
else
{
system.out.println(“numbers are equal”)
}
```

Question 3:

a) Even, odd

b)
```
switch (num % 2)
{
case 0:
      System.out.println(“even number”);
      break;
Case 1:
       System.out.println(“odd number”);
       break;
}
```

Question 4:

a) int num = (int)(Math.random() * 50) + 1;

b) int num = (int)(Math.random() * 81) + 20;

c) Double num = Math.random() * 10 + 10;

Question 5: (age > 18) should be removed in the second line because if it stays then anyone who is 18 would not fit in any category, For the senior one. It should just be ‘else’ not ‘else if (age > 65)’ otherwise 65 wont fit in any category

Question 6:

a) True
b) False
c) True
d) True
e) True
f) True
g) True

Question 8:

a) True
b) False, a nested if places an if inside another statement. However an if else if checks multiple conditions in a sequence
c) False, switch statements don't support a double expression
d) True
e) False, (int) is used for integers
f) True
g) True
h) False, the ! has more priority than the &&
i) True
j) False, missing 'math'